

Element A

The problem is clearly and objectively identified and defined with considerable depth, and it is well elaborated with specific detail; the justification of the problem highlights the concerns of many primary stakeholders and is based on comprehensive, timely, and consistently credible sources; it offers consistently objective detail from which multiple measurable design requirements can be determined.

The Brentwood High School Theatre and Production Team is performing the live action Beauty and the Beast play in March of 2024. As a part of the production team, most of the props used in the play are engineered and designed by the BHS production team. These props built by our teams would be the Magic Mirror and the Enchanted Rose. In this case, we are tasked with working on the Magic Mirror. In the Beauty and the Beast play, Belle holds up the mirror and sees a vision through it. When she holds up the mirror to her face, there is a bright light that shines out from the mirror that reflects off her face and is also visible to the audience. The design and creation of this magic mirror was given as a task for our Engineering class. Our primary stakeholder is Ms. Moody, who is the director of the Beauty and the Beast play at BHS. Other primary stakeholders include, but is not limited to, the BHS Theatre and Production Crew, the members of the cast, the engineering class, and the audience viewing the play. Through conducting multiple interviews with Ms. Moody, we identified the most important criteria of a successful mirror project are the visibility of the light projection coming from the mirror, the overall style and design, and the comfortability of the mirror in the hands of the actors. The main actors that will be holding the mirror are Belle, Gaston, and the Beast. At each point in the play that they are holding the mirror they will not be holding the mirror for more than 20 seconds. At

some point, those actors will also be wearing gloves. As a result of these gloves affecting the comfortability of the mirror in their hands we also acquired these gloves to access the comfortability of the mirror handle with the gloves for less than 20 seconds. As measurable criteria, we identified that the ideal size of the mirror handle is that it is not larger than $1\frac{1}{4}$ inches in diameter and no smaller than $\frac{3}{4}$ inch diameter. We also identified that the mirror should weigh no more than 5 pounds and no less than 1 pound, should be no larger than 10 inches by 10 inches and no smaller than 5 inches by 5 inches, and should also be an elliptical shape. Through testing lights on the actual stage and gauging the visibility of them from various angles and distances in the auditorium, we also decided that the ideal type of light are RGB LED lights. Through the completion of these criteria along with the approval of the aesthetic from Ms. Moody, the Beauty and the Beast Mirror Project will be successful.